Kinetic and Potential Energy Questions Name: _____

1. A model glider has a mass of 1 kg. How much potential energy does it have 2 meters off the ground?

2. The same model has a velocity of 2.2 m/s. How much kinetic energy does it have?

3. If the same model descends 2 meters and all it's potential energy is converted to kinetic energy, what is the glider's change in velocity?

4. A full-sized glider has a weight of 4,900 N, while it's pilot has a weight of 825 N. If it is 1,000 meters off the ground, how much potential energy do the plane and pilot have?

5. The same glider from Problem 4 has a velocity of 35 m/s. How much kinetic energy does it have?

6. The same glider from Problem 4 has a velocity of 35 m/s. The glider descends 900 meters. What is it's new velocity?

7. Kinetic energy can be defined as....

8. The equation to be used to calculate kinetic energy (Ek) is...

9. The SI unit in which Ek is often measured is the ..._____.

10. Rodger Maris swung a bat which had a mass of 2 Kg at a velocity of 45 m/s.
How many joules of kinetic energy could he give to a ball?

11. Barry Bonds swings a bat which has a mass of 1.5 Kg at a velocity of 55 m/s.
How many joules of kinetic energy could he give to a ball?

12. Which is more important to hitting a home run - a heavier bat or a faster swing?

13. A golf pro swings his driver which weighs .75 kg at a velocity of 60 m/s.
Calculate the kinetic energy of the club.

14. Calculate the Ek of a car which has a mass of 1000 kg and is moving at the rate of 20 m/s.

15. What is the Ek of a soccer ball which has a mass of 0.8 kg and is kicked at a velocity of 10 m/s?

16. Calculate the Ek of a running back that has a mass of 80 kg and is running at a velocity of 8 m/s.